



Description

Carried out to simulate the effect of a specified load.

Simulation of FEA

Date: 21 October 2023
Designer: Group 4
Study name: Backrest Test
Analysis type: Static

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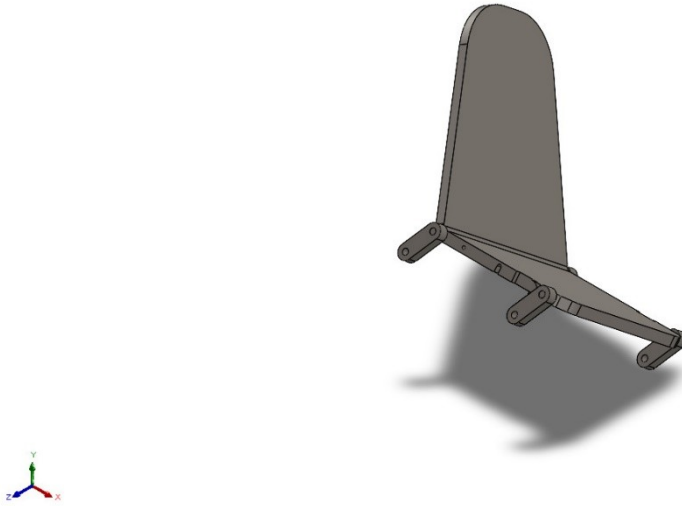
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Assumptions

The pins and the seat are fixed.

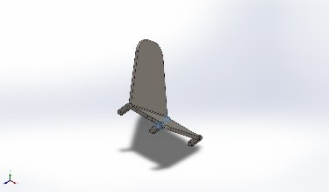
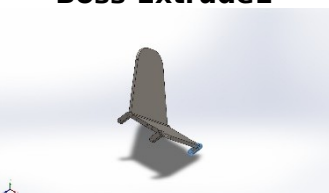




Model name: FEA

Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Boss-Extrude1 	Solid Body	Mass:0.351222 kg Volume:4.56133e-05 m ³ Density:7,700 kg/m ³ Weight:3.44198 N	C:\Users\Ibrahim\ Downloads\ assemb_final\ link.SLDPRT Oct 21 00:15:59 2023
Boss-Extrude1 	Solid Body	Mass:0.351222 kg Volume:4.56133e-05 m ³ Density:7,700 kg/m ³ Weight:3.44198 N	C:\Users\Ibrahim\ Downloads\ assemb_final\ link.SLDPRT Oct 21 00:15:59 2023
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SOLIDWORKS

Analyzed with SOLIDWORKS
Simulation

Simulation of FEA

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Study Properties

Study name	Backrest Test
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\Ibrahim\Downloads\assemb_final)

Units

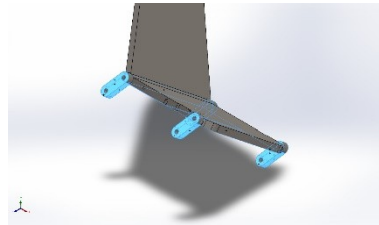
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

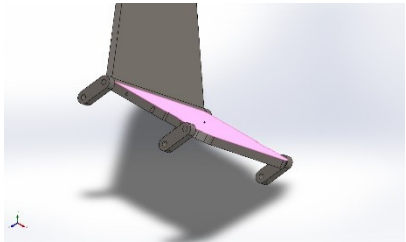
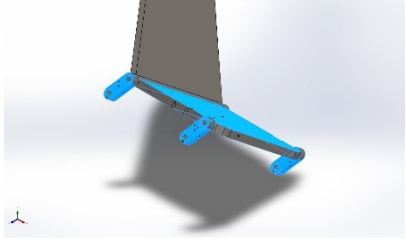


Material Properties

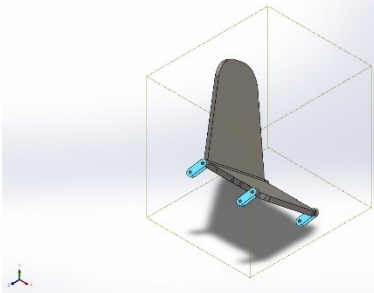
Model Reference	Properties	Components
	Name: Alloy Steel (SS) Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 6.20422e+08 N/m ² Tensile strength: 7.23826e+08 N/m ² Elastic modulus: 2.1e+11 N/m ² Poisson's ratio: 0.28 Mass density: 7,700 kg/m ³ Shear modulus: 7.9e+10 N/m ² Thermal expansion coefficient: 1.3e-05 /Kelvin	SolidBody 1(Boss-Extrude1)(link-1), SolidBody 1(Boss-Extrude1)(link-2), SolidBody 1(Boss-Extrude1)(link-3), SolidBody 1(Boss-Extrude1)(link-4), SolidBody 1(Cut-Extrude7)(seat-bottom-1)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details			
Fixed-1		Entities: 21 face(s) Type: Fixed Geometry			
Resultant Forces					
Components		X	Y	Z	Resultant
Reaction force(N)		-42.0072	-99.4746	107.397	152.296
Reaction Moment(N.m)		0	0	0	0

Load name	Load Image	Load Details		
Gravity-1		Reference: Face< 1 > Values: 0 0 -9.81 Units: m/s ²		
Force-1		Entities: 148 face(s) Type: Apply normal force Value: 750 N		

Interaction Information

Interaction	Interaction Image	Interaction Properties
Global Interaction		Type: Bonded Components: 1 component(s) Options: Independent mesh

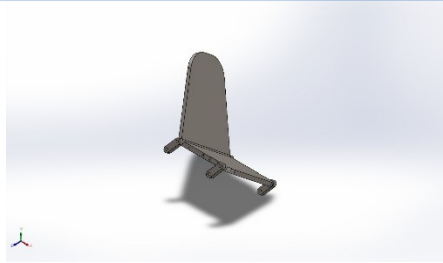
Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	31.9239 mm
Minimum element size	2.2961 mm
Mesh Quality	High
Remesh failed parts independently	Off

Mesh information - Details

Total Nodes	32155
Total Elements	17660
Maximum Aspect Ratio	24.265
% of elements with Aspect Ratio < 3	82.5
Percentage of elements with Aspect Ratio > 10	0.708
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:09
Computer name:	IBRAHIM-PC

Sensor Details

Sensor name	Location	Sensor Details
Mass1		Value : 4,064.61 g Entities :5 component(s) Result :Displacement Component :URES: Resultant Displacement Criterion :Model Max Step Criterion : Across all Steps Step No.:1 Alert Value: NA
Displacement1		Value : 1.787e-02 mm Entities : Result :Displacement Component :URES: Resultant Displacement Criterion :Model Max Step Criterion : Across all Steps Step No.:1 Alert Value: NA

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-42.0072	-99.4746	107.397	152.296

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

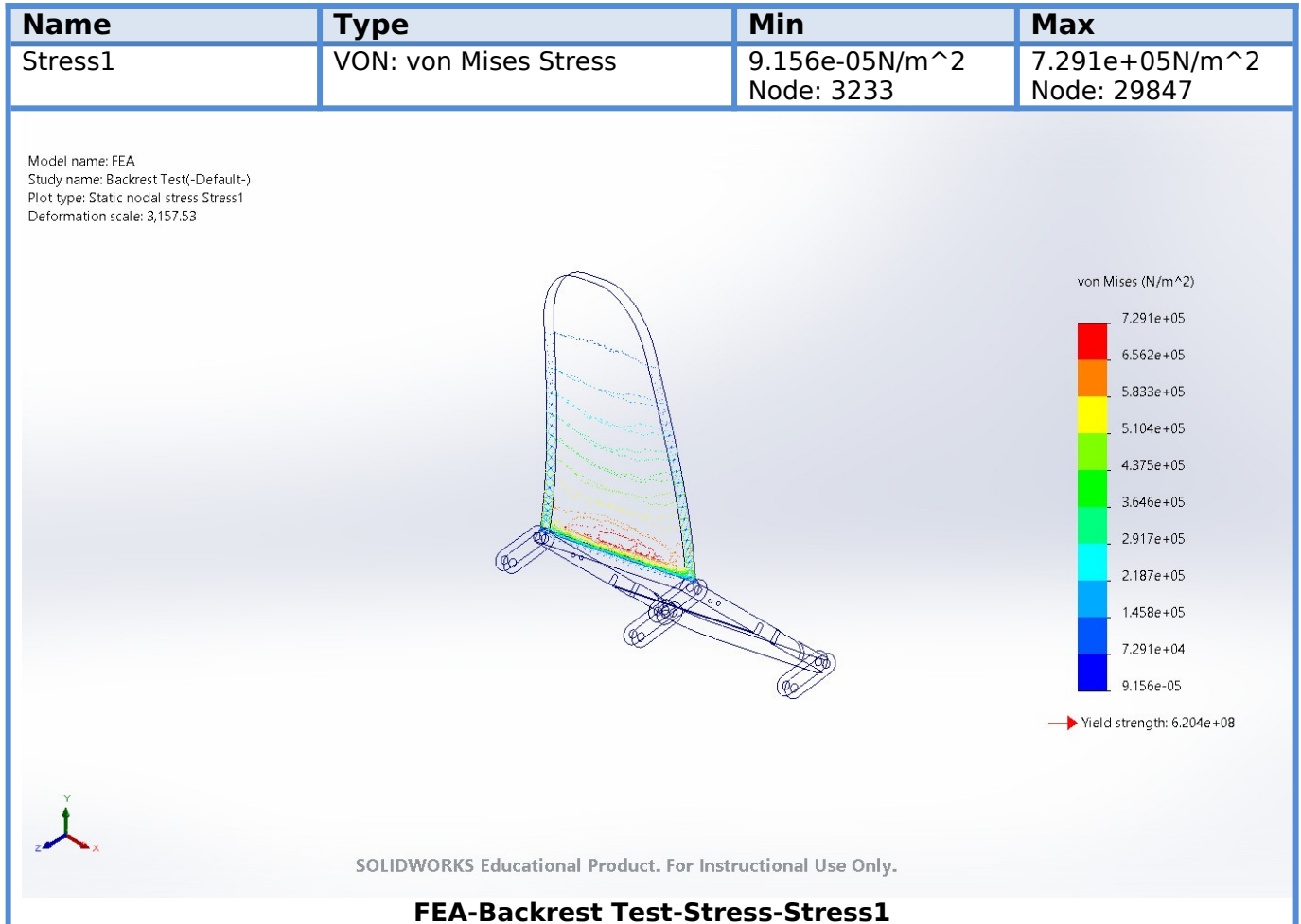
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0.0267402	165.905	-130.974	211.373

Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

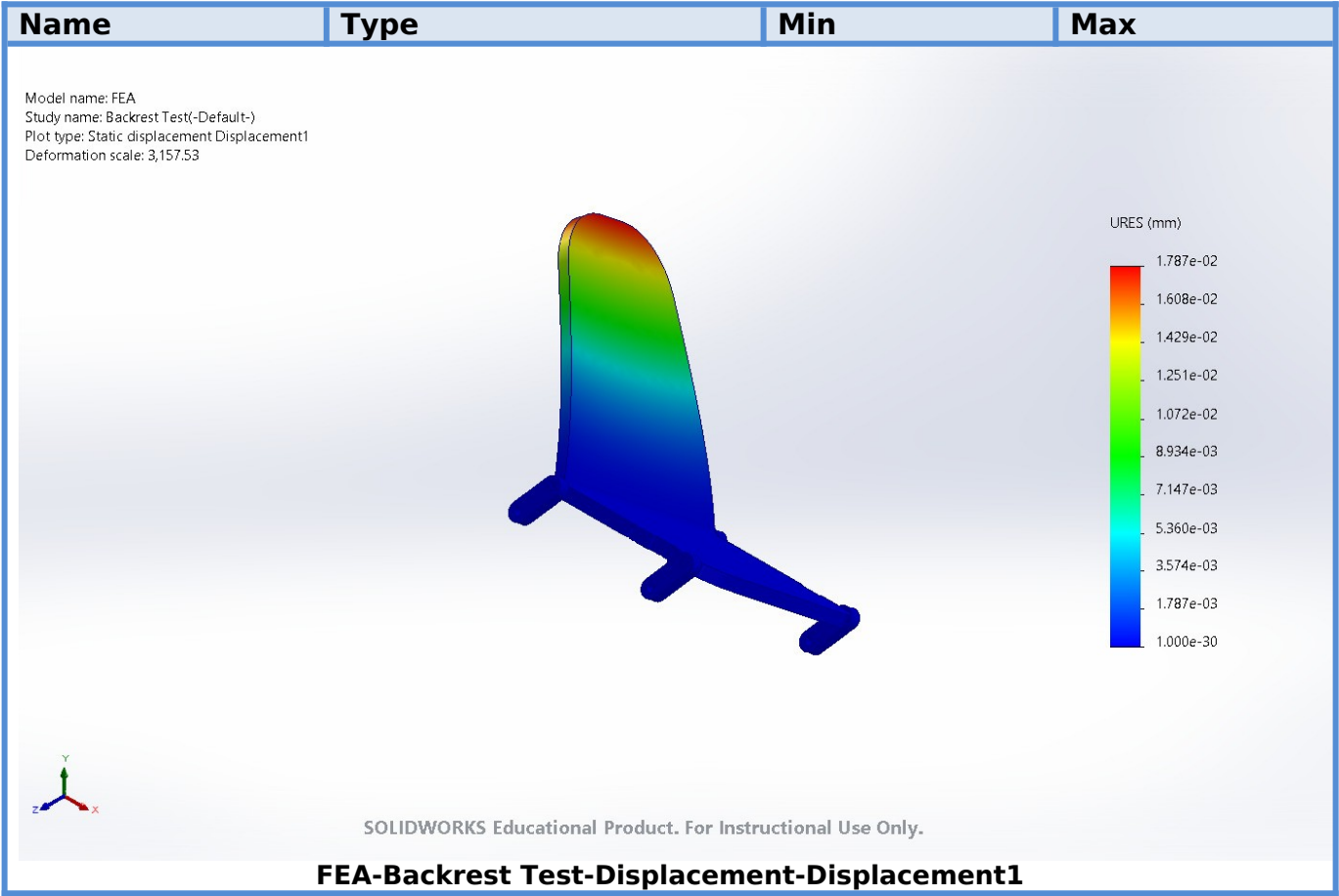


Study Results



Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 1	1.787e-02mm Node: 9562





Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	1.539e-10 Element: 16	2.019e-06 Element: 16745

